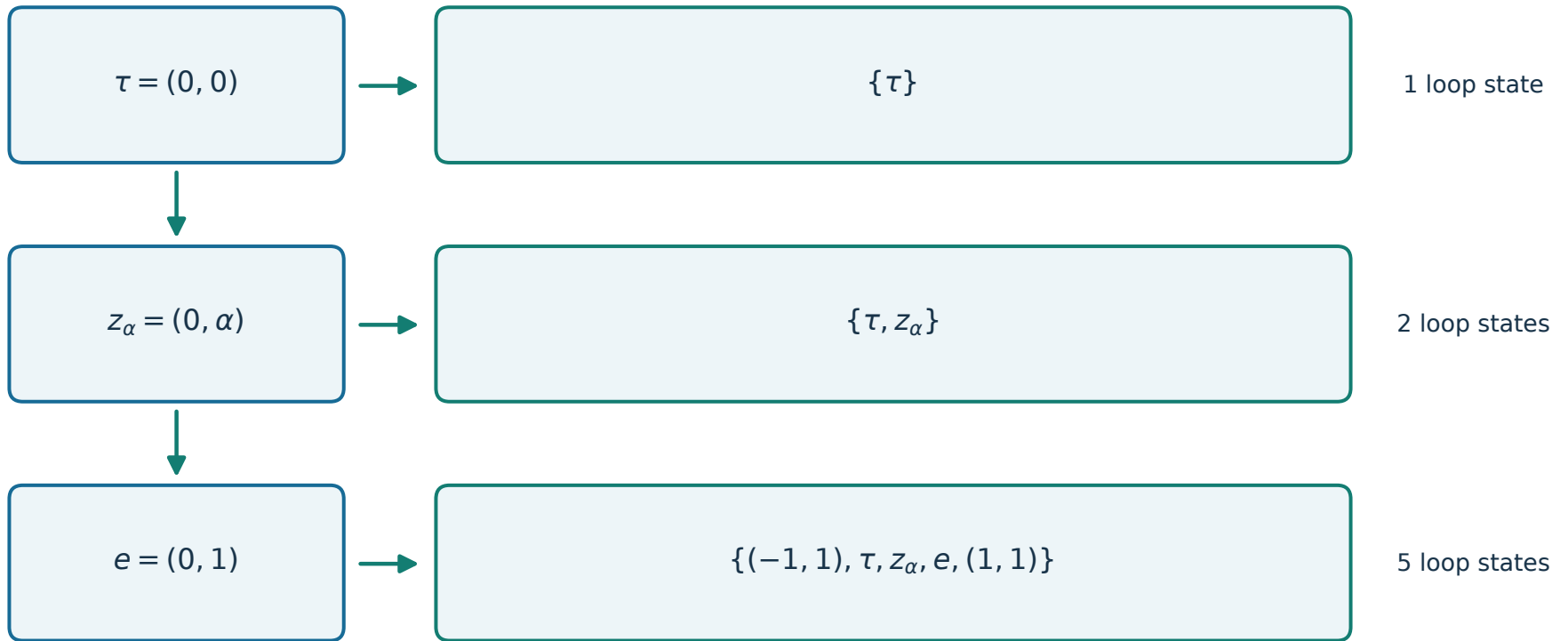


Loops retain their support basepoint

Exact example: $L = \{0 < \alpha < 1\}$, $D = 0$, $\Lambda_D = \mathbb{Z}$, $e^a = 1$, one coordinate.

All one-spheres, before any amplitude or support is discarded



Each state is a distinct elementary homotopy class.

The complete zero-stage relation

$$(b, \nu) \mathcal{R} (b', \mu) \iff \nu \leq \mu \quad \text{and} \quad d_{\mathbb{R}/\mathbb{Z}}(b, b') \leq 1$$

Support may increase along an edge. The original distance bound remains.